Heaven's Light is Our Guide

Dept. of Electrical & Computer Engineering (ECE),
Rajshahi University of Engineering & Technology

Course Code: ECE 2215

Course Title: Data Base Systems

Instructed by:
Mst. Nafia Islam Shishir
Lecturer, ECE, RUET

Instructor's Details

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Designation: Lecturer

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Course Outcomes (COs)

COs	Description	Taxonomy domain/level	POs	K	P	A
CO1	**Describe** the basic concepts and applications of database systems.	Cognitive/ Remember	1			
CO2	**Apply** MYSQL code to implement database system.	Cognitive/ Apply	2			
CO3	**Illustrate** the E-R Diagram, transaction concurrency control and recovery system.	Cognitive/ Analyze	1			

Program Outcome (POs)

PO1 – Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4, respectively, to develop solutions of complex engineering problems.

PO2 – Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).

Teaching Learning Method

COs	Teaching-Learning strategy	Assessment strategy
CO1	Lecture, Practice and Discussion	Class Test, Assignment
CO2	Lecture, Practice and Discussion	Class Test, Assignment
CO3	Lecture, Practice and Discussion	Class Test, Assignment

Assessment Tools

Assessment Tools		Marks (%)	
Continuous Assessment (CA)	Lab Participation	10%	35%
	Continuous Assessment	10%	
	Project	15%	
Summative Assessment (SA)	Quiz	20%	65%
	Lab Final	20%	
	Board Viva	25%	
	Total	**100%**	

Course Content

Lecture No.	Name of the Topics
1	Introduction to Course Plan, introduction to database
2	Files and Database, Database Management system & Applications
3	Structure of DBMS
4	Structure of Relational Databases
5	Relational algebra, Relational algebra operations
6	Relational algebra, Relational algebra operations
7	Modification of the database, Pitfalls in relational database design
8	Data Definition Language, Data Manipulation Language, Basics of SQL
9	Query designing in SQL using aggregate functions and nested queries
10	CT 1

Course Content

11	Query designing in SQL using aggregate functions and nested queries
12	Embedded SQL, Triggers, Procedures
13	Entity types, Entity set, Attribute and Key
14	Relationships, Relation types, Entity relationship, ER modeling, ER diagrams
15	Relationships, Relation types, Entity relationship, ER modeling, ER diagrams
16	Database design using ER diagrams
17	CT 2
18	Normal forms
19	NORMALIZED Relations and Database performance
20	De-normalization

Course Content

20	Transaction Management
21	Transaction Management
22	Concurrency Control, Locked Based protocols
23	CT3
24	Deadlocks
25	Recovery System, Failure Classifications
26	Recovery and atomicity
27	No SQL System
28	Distributed Systems
29	CT 4
30	Review

Suggested Books

1. **Database System Concepts, Seventh Edition,**
   - by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan

2. SQL, PL/SQL: The programming language of Oracle
   - by Ivan Bayross

2. Database Management Systems
   - by Raghu Ramakrishnan and Johannes Gehrke

Course Conducting Policies

1. It is the student's responsibility to gather information about the assignments and covered topics if he/she miss the lecture.
2. Regular class attendance is mandatory. Points will be taken off for missing classes.
3. Without **50% of** attendance, sitting for the final exam is **NOT allowed**.
4. The students must enter the **classroom on time** to get attendance. **No student** will be allowed to enter the classroom after attendance has been taken.

Course Conducting Policies

5. Once the attendance is done, a student can leave the class if he or she thinks that he or she is not getting benefits from the class.

6. Students are encouraged **to participate** in the class discussion and to ask questions. The student can ask any question without any hesitation as long as he or she can't understand the topics being discussed; please keep in mind that if you don't understand**, it's not your fault, it's my limitatio**n that I could not make you understand. The class is expected to be interactive.

7. The date and syllabus of quiz/class test will be announced beforehand.

Course Conducting Policies

8. Students will be **notified** in due time for class cancellation, extra class, make-up class and tutorial class.
9. It is expected that the student will also provide some new knowledge related to the curriculum and then make the class a **place of knowledge sharing among all participants, both teacher and students**.
10. Any attempt at **unfair means** in the examination is **strictly prohibited**.

Thank You

If you have any queries, you can ask.